

Test-4

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1) → ~~What~~ Data is a collection of raw facts, values or information that can be stored and processed.

2) → ~~What is~~ Database is an organized collection of data that can be stored, accessed, managed and update efficiently.

3) → Database Management System

4) → Relational Database Management System

5) → ~~What is~~ MySQL?

→ MySQL is an open source-relational database management system (RDBMS) that uses SQL to create, manipulate and manage

6) → What is SQL?

→ - SQL - Structured Query Language
- SQL is a language used to communicate with relational Database.

7) → Name Any two Database Model?

- 1) Hierarchical Database
- 2) Relational Database

8) What is row in database models?

→ row is A single entry in table

9) What is purpose of the DISTINCT keyword?

→ It will return unique value

10) What does the % wildcard represented in MySQL LIKE?

→ % - percentage symbol represent zero or multiple character

Section B

11) Difference between Database and DBMS

Database

DBMS

- | | |
|--|---|
| 1) organized collection of data. | 1) Database Management System |
| 2) Can be stored, accessed, managed and update efficiently | 2) Can support different database model depending on the system |
| 3) Fast Searching | 3) general database management system. |

12) Differentiate between SQL and MySQL

SQL MySQL

- | | |
|------------------------|--------------------------------|
| 1) Language | 1) RDBMS / software |
| 2) Used to write query | 2) Manages Relational Database |
| 3) contain SQL Command | 3) Execute SQL Command |

14) Difference between IN and BETWEEN operators?

IN BETWEEN

- | | |
|------------------------|----------------------|
| 1) To specify multiple | 1) BETWEEN a certain |
|------------------------|----------------------|

possible values for Range
a column

- 2) Select Data with IN 2) To give a range
in mySQL we use
Between operators

15) Explain IS NULL and NOT NULL with Suitable
SQL examples.

→ 1) IS NULL

select data is null;
where age is null;

e.g - MySQL select * from employee

→ where address is null;

2) NOT NULL

e.g - select * from employee where age is not
null;

13) write any four SQL command categories
and give one command for each.

- 1) DDL - Data Definition language
2) DML - Data Manipulation Language
3) DQL - Data Query Language
4) DCL - Data Control Language

Section C

16) Explain DBMS and RDBMS at least 3

→ 1) DBMS

- Database Management System
- General Database Management system
- Can support different database model depending on the system.

2) RDBMS

- Relational Database Management system
- Based on relational model
- Data represented using table

17) Explain the following database models with their basic structure.

→ 1] Hierarchical Database

- A hierarchical database organizes data in tree-like structure

parent
|

+ --- child

2] Network Database

- A network database organizes data as interconnected records and supports

multiple relationships among records.

Student

Course Teacher

3] Relational Database

— A relational database organizes data into tables (relations) consisting rows and columns.

Student_id name Age Course_id

101	Shrawani	21	CO1
102	Siddhi	22	CO2

4] Object-oriented Database

— An object-oriented database stores data in the form of objects.

e.g:—

Student object

+ _____ name = Shrawani

+ _____ age = 21

+ _____ course = B.Tech

5] NoSQL Database

- NoSQL is broad category of database that uses data models other than, or beyond the traditional-table model.

18] Explain the six MySQL constraints given in the notes. Write the purpose of each!

→ 1) NOT NULL

- ~~It~~ a column cannot accept a NULL or empty value.

2) Unique

- All values in a column are completely distinct from one another

3) DEFAULT

- All values entered into a column satisfy a specific Boolean condition.

4) Check

- All values entered into a column satisfy a specific Boolean condition.

5) primary key

— uniquely identifies each record in a table.

6) primary key

— A column that links to a primary key in another table.